Benjamin van Duin

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EDUCATION

University of North Carolina at Chapel Hill *May 2027*

Bachelor of Science: Environmental Science – Ecology and Natural Resources Concentration

Bachelor of Arts: Geography

GPA: 3.95

National Residence Hall Honorary

East Chapel Hill High School *June 2023*

National Honor Society | National Latin Honor Society

PROJECTS

Coastline Change Analysis, Mekong Delta, Vietnam — *Google Earth Engine, remote sensing*

- Quantified 36 years (1988–2024) of coastline change using Landsat 5/8 imagery and unsupervised clustering in Google Earth Engine, finding a net land loss of ~72 km² (350 km² erosion vs. 272 km² accretion) consistent with published Can Tho University estimates.
- Linked observed erosion to a measured decline in suspended sediment concentration, supporting hypothesized dam-driven disruption of delta sediment supply.

Species Distribution Model, Carolina Wren — *R/RStudio, GBIF, TerraClimate*

- Built a species distribution model using GBIF presence data and temperature/precipitation covariates, fitting a binomial logistic regression (AUC = 0.71) to predict habitat suitability.
- Validated predictions against a decade of eBird abundance data, finding suitability correlated with temperature change (R^2 = 0.50) but not with observed abundance trends.

Population Exposure to PM2.5, European NUTS3 Regions — *R/RStudio, spatial statistics*

- Mapped PM2.5 exposure across ~1,300 European regions using EEA and WorldPop data, finding ~28.7 million people living above WHO air quality guidelines.
- Applied spatial error regression and LISA clustering to test population density as a predictor, finding strong spatial clustering (Moran's I = 0.86) but that density explains under 1% of variance.

WORK EXPERIENCE

Resident Advisor Mentor, Carolina Housing, Chapel Hill, NC *August 2026 - present*

- Supervise and mentor a team of Resident Advisors, overseeing implementation of community programming, conflict mediation, and emergency response protocols across the residence hall.
- Retain full RA responsibilities — resident safety, community programming, and conflict mediation — while adding team leadership and staff development duties.
- Manage a program budget and P-card covering RA programming, staff development, recognition, and training, ensuring expenditures comply with university procurement policies.

Resident Advisor, Carolina Housing, Chapel Hill, NC *August 2024 - May 2026*

- Monitored the safety and security of approximately 190 residents in Hardin Hall by conducting regular inspections, responding to emergencies, and fostering a culture of accountability and community awareness.
- Designed and implemented engaging community programs that fostered meaningful connections between residents, while promoting resilience, values development, and a commitment to equity and inclusion across the community.
- Mediated challenging roommate conflicts and guided residents in developing collaborative yearly living agreements to promote harmony and accountability.
- Developed creative visual content as part of the community engagement plan to communicate important information and enhance resident interaction.

Research Assistant, The Wildlife Biogeography Lab, Chapel Hill, NC *April 2025 - present*

- Identified mammal species from camera trap photographs and gave them a species “tag” in their metadata using the open source software Digikam.

- Participated in numerous trips to research sites near the Albemarle Sound to deploy bucket camera traps, standing camera traps, and collect previously deployed cameras.

FANgineer Leader, Carolina Athletic Hospitality, Chapel Hill, NC *August 2024 - November 2024*

- Oversaw a team of FANgineers while continuing to perform the typical FANgineer duties.
- Responsible for direct communication with the Gameday Interns and Senior Manager using a two-way radio to ensure proper management of emergencies and issues.

FANgineer, Carolina Athletic Hospitality, Chapel Hill, NC *August 2022 - May 2024*

- Enforced operational policies in premium seating areas in Kenan Stadium, the Dean Smith Center, and Boshamer Stadium to ensure a smooth gameday experience.
- Protected the exclusivity of premium seating areas by scanning tickets and providing wristbands promptly in high-volume settings.
- Served as a liaison between donors, caterers, maintenance staff, custodial staff, CAH Directors, and Athletic Department personnel through efficient communication.

Marketing Intern, Carolina Dining Services, Chapel Hill, NC *June 2024 - August 2024*

- Devised, filmed, and edited short-form content to promote the brand and its locations on social media platforms.
- Presented on various aspects of the dining halls and meal plans to groups of approximately 50 first-year students and parents to improve sales and brand visibility.
- Arranged tables and displays to bolster engagement with students and parents through direct interactions.

SKILLS

Technology: Microsoft Office Suite, Adobe Creative Suite, Google Earth Engine, QGIS, R/RStudio
Languages: Fluent in Dutch